MARS MISSION DESIGN LAB / STUDENT PORTFOLIO

Student Mars Mission Portfolio

Student / team: Mars Engineering Team

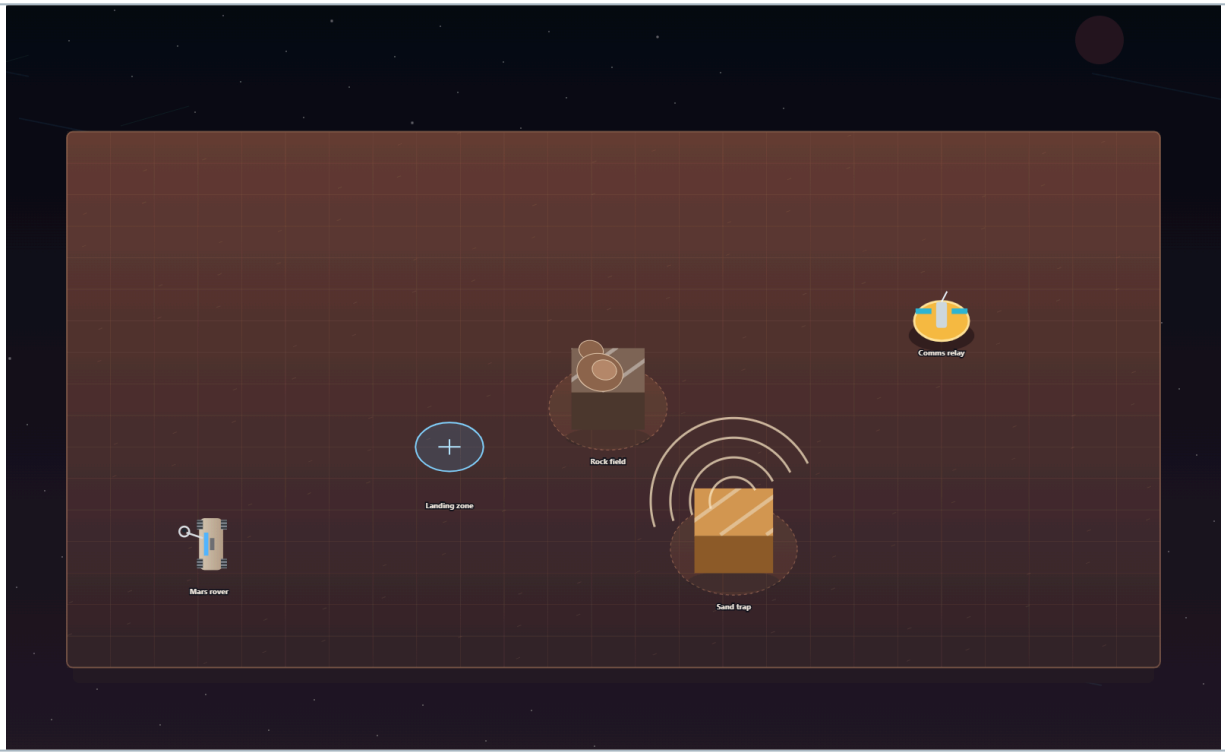
Mission challenge

Level 3 · Sample Return Engineer
Collect two samples, relay the findings, and deliver them to a cache. Include at least 3 obstacles to avoid.

Engineering rank

Engineering Cadet · 0/4 levels passed in this browser
This design still needs a successful Python demonstration.

Mission design screenshot



Design captured for this proposal. Dashed line: planned route. Solid line: latest coded traverse.

Mission resources and testing

> Build credits: 510 / 1,000 | Terrain obstacles: 2 / 3 minimum
> Planned route: 0 map units | Planned energy: 0 / 100
> Latest test: Awaiting code test

Our science question

How do rock samples differ across our imagined Mars science sites?

Our engineering decisions

We placed three terrain obstacles between our science sites. We chose a route with space around each hazard. After testing, we revised our travel times and explained each science action.

Our Python rover program

Student-written program, including comments and indentation.

```
# Student-written rover program
speed = 60
for step in range(2):
  roll(speed, 90, 0.5)
  turn(90)
# We will add the science actions for our own layout.
collect()
flash_colour_code("orange-blue")
stop()
# END OF STUDENT PROGRAM
```

Design checklist (code not yet demonstrated)

STILL NEEDED: Collect two different rock samples
STILL NEEDED: Visit a communications relay
STILL NEEDED: Finish at a sample cache
STILL NEEDED: Include at least 3 terrain obstacles (2/3 placed)
STILL NEEDED: Keep the route clear of hazards
PASS: Stay within build and energy limits
STILL NEEDED: Demonstrate your mission with a successful Python run

Placed equipment and obstacles

Object	Type	Position (X, Y)	Credits
Mars rover	Mars rover	130, 540	300
Landing zone	Landing zone	350, 400	80
Comms relay	Comms relay	800, 250	130
Rock field	Rock field	495, 350	0
Sand trap	Sand trap	610, 530	0

Planned flight path

No planned waypoints. The Python program controls the rover directly.

Independent classroom simulation. Imagined Mars terrain and simplified build credits, energy and mission actions.